AI Programming With Python

Mid Term

Submission Instructions

- Watch the assignment instruction video carefully before starting.
- Create a Google Colaboratory (.ipynb) file in your Google Drive.
- Add a **Text Cell** with:
    - Your registered email address (used for your Phitron account).
- Write all answers in a **single notebook file**.
- For each question:
    - Add a **Text Cell (question number)**
    - Add a **Code Cell (solution)**
- Display all outputs clearly for evaluation.
- Share the file with **“Anyone with the link”** → **Viewer access**.
- Submit only the **Colab link**.

Note: Each question carries **10 marks**.

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0) Starter Code (Run First)

```
# You must run this code and use this dataset for all questions

import numpy as np
import pandas as pd

np.random.seed(24)

num_employees = 50  # you can increase this if needed

# Employee data
emp_ids = np.arange(101, 101 + num_employees)
emp_ages = np.random.randint(20, 55, num_employees)
emp_income = np.random.randint(35000, 95000, num_employees)

# Fixed set of roles (repetition allowed)
roles_list = [
    "HR Manager",
    "Software Developer",
    "Accountant",
    "Marketing Executive",
    "Sales Executive",
```

```
    "Backend Developer",
    "HR Executive",
    "Financial Analyst",
    "Digital Marketer",
    "Operations Manager"
]

emp_roles = np.random.choice(roles_list, num_employees)

# Create DataFrame
data = pd.DataFrame({
    "emp_id": emp_ids,
    "emp_age": emp_ages,
    "emp_salary": emp_income,
    "role": emp_roles
})

# Save dataset
data.to_csv("staff_data.csv", index=False)
```

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1) NumPy — 5 Questions

Q1.

Use NumPy arrays , the `emp_ages` and `emp_income`.
Print both arrays and display:

- data type
- number of dimensions
- shape
- total number of elements

Q2.

Using NumPy functions:

- Find the **maximum and minimum salary**
- Calculate the **average salary**
- Calculate the **average age**
  Print all results.

Q3.

From the `emp_ages` array:

Extract employees who:

- **are aged between 25 and 40,**
  AND

Emp_salary : have salary greater than 50,000

Then :

- Print the filtered values
- Count how many employees satisfy this condition

Q4.

Create a new NumPy array where each employee's age is adjusted based on:

- Add **4 years** if age is less than the average age
- Add **2 years** if age is greater than or equal to the average age

Do NOT modify the original array.

Print:

- original array
- updated array
- average age used for condition

Q5.

Using NumPy:

- Assume all salaries of **Tech roles** increase by **8%**
- Calculate the **new total salary expense**
- Find the **overall increase in expense** after the raise

Print both old and new totals along with the difference.

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2) Pandas — 5 Questions

Q6.

Load the `staff_data.csv` file into a Pandas DataFrame.

- Display the **first 3 rows**
- Show dataset information using `.info()`
- Display summary statistics using `.describe()`

Q7.

Filter the dataset to include only:

- `emp_id`, `emp_age`, `role`

Then:

- Select employees whose **age is between 25 and 40**
- Sort them by **emp_id**
- Display the **last 3 rows** of the result

Q8.

Filter the DataFrame to show only employees working in the **"HR" role**.

- Print the filtered data
- Show the total number of HR employees

Q9.

Sort the DataFrame by:

- `emp_age` (ascending)
- and then by `emp_salary` (descending)

Display the **top 5 employees** after sorting, including:

- `emp_id`
- `role`
- `emp_salary`

Q10.

Using `.loc`:

- Increase salary by **12% for "Software Developer" and "Backend Developer"**
- Increase salary by **5% for all other roles**

Then:

- Print the dataframe that includes **role , previous_salary , new_salary , increase_of_salary**